



V93000 Specifications

Pin Scale 1600 Digital Card

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A warning calls attention to a procedure, practice or similar process, which, if not correctly performed or adhered to, could result in death, injury or harm to personnel.

➡ Do not proceed beyond a warning unless the indicated conditions are fully understood and met.

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A caution calls attention to a procedure, practice or similar process, which, if not correctly performed or adhered to, could result in damage to or destruction of part or all of the test system equipment.

➡ Do not proceed beyond a caution unless the indicated conditions are fully understood and met.

PS1600 specifications

Note

The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).



Pin Scale 1600 Digital Card

Further information

For more information about the Advantest V93000 SoC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SoC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

PS1600 overview

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test Processor Per-Pin architecture

Channel count	128 channels per card
Parallel vector memory ^[1]	16 MByte (64 MVectors at x4 mode) standard. 32 MByte (128 MVectors at x4 mode) optional (N5867U). 64 MByte (256 MVectors at x4 mode) optional (N5868U). 112 MByte (448 MVectors at x4 mode) optional (N5869U). 224 MByte (896 MVectors at x4 mode) optional (N5870U) for the PS 1600B revision.
Maximum scan depth ^[2]	57 GVectors (at x4 mode) for PS1600A 114GVectors (at x4 mode) for PS1600B
PRBS per Pin	PRBS up to 2^{32} , programmable seed and compare User defined compliance and serial standard pattern space (32k per pin)
SmartLoop™	Active, retimed, differential & single ended, loop direction changeable

Speed

Maximum I/O data rate at ≤ 1.8 V (pattern, PRBS, SmartLoop™)	1600 Mbps
Maximum I/O data rate at ≤ 3 V (pattern, PRBS, SmartLoop™)	1200 Mbps
Maximum clock rate at ≤ 1.8 V	800 MHz

^[1] Vector memory is used for pattern, setup data and result capture

^[2] Maximum scan depth support total of 16 chains per instrument, 7.168GVectors max (drive only, single bit mode) or 3.584 GVectors in x4 mode for EACH chain. Total data volume per instrument ~ 57GVectors (in x4mode). For PS1600B: 14.336 GVectors max (drive only, single bit mode) or 7.168 GVectors in x4 mode for EACH chain.

Maximum clock rate at ≤ 3 V	600 MHz
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Maximum external input voltage (compliance) range

for pogo pins in standard voltage mode	-3.0 V to +8.0 V
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Test programming and debugging

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment, and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test methods. User-specific tests are programmed with test methods in C++. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

Compatibility

This digital card is supported starting with SmarTest 7.1 It is a new member of the V93000 scalable platform expanding the digital card portfolio. It can be used with Pin Scale 9G, DC Scale, DPS, MBAV8/MBAV8+ and Port Scale RF cards. Configuration recommendations are available to optimize fleet utilization..

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period ^[3]	3 months

Environmental

Operating	15°C to 30°C (59°F to 86°F) ^[4]
Specification warranted temperature	20°C to 30°C (68°F to 86°F) ^[4]
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

License overview

N8030BC-PRM: Pin Scale 1600A/B Base Capability License (128 pins, 100Mbps, 16MB), perpetual, per board, floating.

100Mbps speed is supported as standard. Faster speeds require upgrade licenses (N5862UPRM - N5864UPRM).

For an overview of all licensed V93000 products by Product ID, as they are referenced on the Entitlement Certificates, see "Licensed V93000 products (Topic 130665)".

^[3] Valid at ambient temperature within ± 2.5 K of calibration temperature. For temperature requirements during calibration please see "Maintenance Guide".

^[4] For E8080E limited up to 25°C.

PS1600 Timing

PS1600 OTA and EPA

System OTA and EPA

Overall timing accuracy (t_{OTA})	< ± 200 ps
Edge placement accuracy ($t_{EPA}=t_{OTA}/2$)	< ± 100 ps

Per-board OTA and EPA

Overall timing accuracy per board (t_{OTA})	< ± 160 ps
Edge placement accuracy per board ($t_{EPA}=t_{OTA}/2$)	< ± 80 ps

Edge placement

Edge placement resolution	1 ps
Edge placement range	-4 to 32 sequencer periods Edge placement range maximal -6300 ns to +19000 ns. For sequencer period >752 ns the edge placement range is -4 to 12 sequencer periods

PS1600 vector period

Vector period range	2.5 ns to 31250 ns
Vector period accuracy	± 15 ppm of period setting
Vector period resolution	0.001 fs

PS1600 Clock Data Recovery (CDR)

Clock data recovery (CDR) operates in x4 mode, for example, operates up to 1600 Mbps.

Tracking range	± 7.5 UI
Tracking max. frequency without clock forwarding	33.0 MHz for ± 0.5 UI drift 2.2 MHz for ± 7.5 UI drift
Tracking max frequency with clock forwarding	2.0 MHz for ± 0.5 UI drift 139.0 kHz for ± 7.5 UI drift

PS1600 drive jitter injection

Timing jitter range ^[5]	400 ps peak-to-peak
Jitter frequency range	111 kHz to 94 MHz

^[5] Jitter injection is done with a build-in AWG. Programmable jitter offset range is -280ps to +120ps.

PS1600 driver

AC performance (Verification condition: half the noted voltage into 50 Ohm)

Maximum transition time (10 to 90%)	
at 1.0 V swing	0.6 ns
at 1.8 V swing	0.7 ns
at 3.0 V swing	0.8 ns
Minimum pulse width	
at 1.0 V swing	0.7 ns
at 1.8 V swing	0.8 ns
at 3.0 V swing	0.9 ns

DC performance (high, low, 3rd level, specified into open)

Level range	-1.5 V to 6.5 V
Level resolution	1 mV
Level accuracy	±5 mV
Minimum swing	100 mV ^[6]
Maximum swing	8 V
DC output current	±75 mA

Impedance characteristic

Source impedance	50 Ohm ± 2.5 Ohm
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Z-clamp mode

Voltage range	-2 to 7 V
Maximum voltage window	9 V
Voltage resolution	1 mV
Voltage accuracy	±100 mV ^[7]

^[6] Software allows programming to 0 mV swing, below the specified minimum swing

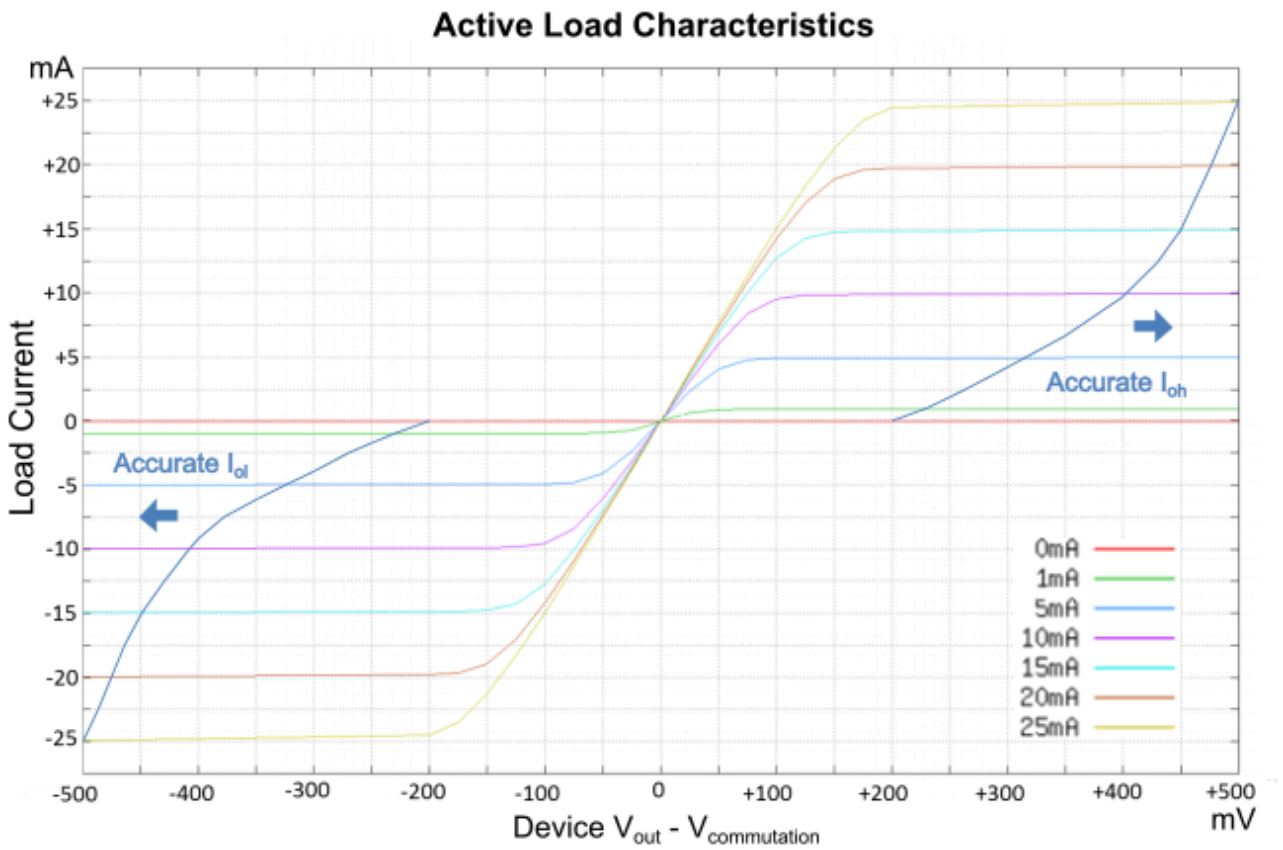
^[7] calibrated at 900µA

PS1600 programmable load

Currents (I_{oh} , I_{ol})	0 to 25 mA
Current resolution	12.5 μ A
Current accuracy	$\pm 75 \mu$ A $\pm 1\%$ of max (I_{oh} , I_{ol})
Commutation voltage range (V_{com})	-1.0 V to 5.5 V ^[8]
Voltage resolution	1 mV
Voltage accuracy	± 100 mV

Sustainable total load may not exceed 1600 mA

Typical characteristics



Active load characteristics

^[8] -1.5 V to 6.5 V if I_{ol} , $I_{oh} < 1$ mA

PS1600 receiver

AC performance

Minimum detectable pulse width (single ended and differential)	600 ps
Intrinsic transition time (10 to 90%)	350 ps, typical

DC performance - single ended

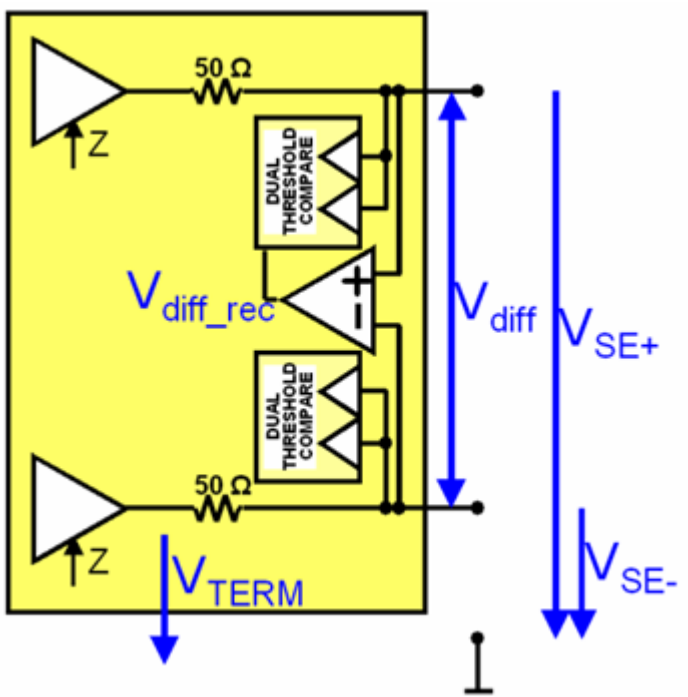
Input voltage and threshold range	-1.5 V to 6.5 V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 uA

DC performance - differential

Input voltage range (V_{SE+} , V_{SE-})	-1.5 V to 6.5 V
Maximum differential input voltage (V_{diff})	±8 V
Differential receive voltage range ($V_{diff_rec} = V_{SE+} - V_{SE-}$)	-1.0V to 1.0V
Threshold voltage resolution	±1 mV
Threshold voltage accuracy	±5 mV
Pin leakage current	±2 uA

Termination

Single ended / center tap differential impedance	50 Ohm ± 2.5 Ohm
Termination level center tap range	-1.5 V to 6.5 V
Termination voltage resolution	±1 mV
Termination voltage accuracy	±5 mV
Switchable pull-down resistor impedance	14.3 kOhm ± 10%



Receiver

PS1600 Parametric Measurement Unit - PMU per pin

PS1600 Voltage force and measure

Maximum Voltage range dependent on selected current range

Current Range	Voltage Range
within range 1 (± 40 mA)	-2.0 V to 5.75 V
within range 2..5 ($\leq \pm 1$ mA)	-2.0 V to 6.50 V

Voltage force

Voltage resolution	200 μ V
Voltage accuracy	$\pm(3 \text{ mV} + I \cdot R)^{[9]}$

Voltage measure (pass/fail and value/SA) using PPMU level comparators

Voltage resolution	1 mV
Voltage accuracy	$\pm(5 \text{ mV} + I \cdot R) \pm 0.5\%$ of reading ^[9]

Value measure using ADCs through PPMU

Voltage resolution	75 μ V
Voltage accuracy in range -2.0 V to 6.5 V ^[10]	± 4 mV
Voltage accuracy in range 0.0 V to 3.3 V ^[10]	± 2 mV
Voltage accuracy of the difference of two voltages measured in range 0.0 V to 3.3 V, measured with different force currents ^{[11] [10]}	± 1 mV

Capacitive loading

Supported PPMU capacitive load	≤ 50 nF or ≥ 10 μ F
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PS1600 Current force and measure

The sustainable total load may not exceed 1600 mA for Pin Scale 1600 (128 channels per card) and 800 mA for Pin Scale 1600-ME (64 channels per card).

The sustainable total load may not exceed 1600 mA.

^[9] I is the actual current, R is the wiring resistance of ≤ 0.5 Ohm

^[10] Requires minimum 16 samples for specified voltage accuracy

^[11] A nominal current I_1 has been forced. The voltage has been measured with the result V_1 . In a subsequent measurement, another nominal current I_2 has been forced. The voltage has been measured with the result V_2 . Both V_1 and V_2 are found within the specified range 0.0 V to 3.3 V. The accuracy of the calculated difference $V_2 - V_1$ is denoted here. This voltage difference shows up with better accuracy as calculating the voltage difference helps to eliminate a systematic measurement error.

Current force

	Range	Resolution	Force accuracy
Range 1	±40 mA	20 µA	±50 µA ±0.2% of setting
Range 2	±1 mA	0.5 µA	±5.0 µA ±0.5% of setting
Range 3	±100 µA	50 nA	±500 nA ±0.5% of setting
Range 4	±10 µA	5 nA	±100 nA ±0.5% of setting
Range 5	±2 µA	1 nA	±40 nA ±0.5% of setting

Current measure using BADC through PPMU

	Range	Resolution	Measure accuracy
Range 1	±40 mA	20 µA ^[12]	±50 µA ±0.5% of reading
Range 2	±1 mA	0.5 µA ^[12]	±1.25 µA ±0.5% of reading
Range 3	±100 µA	50 nA ^[13]	±200 nA ±0.5% of reading
Range 4	±10 µA	5 nA ^[14]	±50 nA ±0.5% of reading
Range 5	±2 µA	1 nA ^[15]	±10 nA ±0.5% of reading

Current measure (value/SA) using PPMU level comparators

	Range	Resolution	Measure typical accuracy
Range 1	±40 mA	20 µA ^[12]	±200 µA
Range 2	±1 mA	0.5 µA ^[12]	±20 µA
Range 3	±100 µA		not supported
Range 4	±10 µA		not supported
Range 5	±2 µA		not supported

Range 1 force current up to ±56 mA for voltages ≤ 3.3 V. Range 1 force accuracy applies.

^[12] Requires minimum 4 samples for specified current accuracy

^[13] Requires minimum 8 samples for specified current accuracy

^[14] Requires minimum 32 samples for specified current accuracy

^[15] Requires minimum 64 samples for specified current accuracy

Current force for differential measurement

Current accuracy of the difference of two currents forced, applied in two consecutive voltage measurements ^[16]	±20 µA ±0.3% of setting in range 1
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Voltage clamp (available in current force)

Voltage range	-2.0 V to 6.5 V
Voltage resolution	1 mV
Voltage accuracy	±100 mV

^[16] A nominal current I_1 has been forced. The voltage has been measured with the result V_1 . In a subsequent measurement, another nominal current I_2 has been forced. The voltage has been measured with the result V_2 . The accuracy of the calculated difference $I_2 - I_1$ is denoted here. This current difference shows up with better accuracy as calculating the current difference helps to eliminate a systematic measurement error.

PS1600 board ADC

The board ADC provides very accurate high impedance voltage measurements through each digital channel. A board ADC is shared among 16 channels.

The PS1600 board ADC allows very high accuracy measurements with 200 μ V relative accuracy by applying the integrated focus calibration^[17]. Here, relative accuracy means that two consecutive measurements are done on the same channel. This is also valid if two consecutive measurements are done on channels out of the group of 16 channels that are sharing one board ADC.

Measurement in standard voltage mode

Voltage range	-3.0 V to 8.0 V
Voltage resolution	75 μ V
Accuracy ^[18]	$\pm$ 1 mV
Relative accuracy between two consecutive measurements on the same channel or channels sharing the same board ADC	Voltage range -0.1 V to 5.1 V with focus calibration^[17] when the difference between measurement results is within 250 mV $\pm$ 100 μ V Voltage range -0.1 V to 5.1 V with focus calibration^[17] $\pm$ 150 μ V Voltage range -2.0 V to 6.5 V with focus calibration^[17] $\pm$ 200 μ V Voltage range -3.0 V to 8.0 V $\pm$ 500 μ V
Input impedance	> 100 MOhm

^[17] Valid for 24h after last calibration with external 3458A Digital multimeter with option 002 from Keysight <= 1 year external calibration interval. Requires >= 4x more averaging as required for standard measurements (see ^[18]).

^[18] Requires minimum 16 samples for specified voltage accuracy

PS1600 Time Measurement Unit - TMU per pin

This topic covers time measurement unit performance data.

DC performance

Specifications of receiver apply (see *PS1600 receiver* on page 10).

AC performance

Maximum input frequency	800 MHz
Linearity error	±5 ps
Accuracy, single timestamp to ARM (time zero)	±35 ps
Accuracy, skew/propagation delay (between different pin resources)	2 * (EPA + 35 ps)
Intrinsic jitter (RMS)	5 ps
Resolution	1 ps

Other (Maximum number of simultaneously active TMUs: 128 shared within slots 1-4 and slots 5-8)

Maximum sampling rate ^[19]	50 Msps
Measurement mode	TMU measurements of single ended or differential compared signals

^[19] The maximum number of samples is limited by the available channel memory.

PS1600 AWG

The PPMU can be used as an arbitrary waveform generator (AWG), supporting both voltage and current waveforms.

All specifications for static PPMU force mode apply.

Max sample rate ^[20]	150 ksps
Max waveform size	only limited by vector memory
Linearity in range of 0...5V (typical values) ^[21]	DNL 250uV (50ppm of range) INL 300uV (60ppm of range)
Linearity in range of 0...3.3V (typical values) ^[21]	DNL 275uV (83ppm of range) INL 300uV (90ppm of range)

^[20] For best AWG performance the neighbor channel of a PS1600 differential pair should not perform any AWG, PMU re-programming or Change Level Operation. PPMU reprogramming may cause noise on adjacent channel of differential pin pair's PPMU. Otherwise max sampling rate may drop to 75ksps and additional timing jitter may be introduced.

^[21] These values are typical numbers, based on generating a 12bit voltage ramp waveform (4096 steps), using the +/- 40mA PMU current range (Range 1). DNL/INL calculation is done using the best fit method.

PS1600 Digitizer

Each Pin Scale 1600 channel can be used as a Digitizer for both voltage and current supporting two use models:

- Vector Based

A command in a pattern triggers a single measurement. Waveforms are captured by adding measurement commands at the appropriate time intervals into a pattern.

- Trigger Based

A single command in a pattern triggers capturing of a complete waveform.

Each use model can be applied to these measurement types:

- High impedance voltage measurement

All specifications for static Board ADC measurements apply.

- Force current, measure voltage

All specifications for PMU per pin apply.

- Force voltage, measure current

All specifications for PMU per pin apply.

Only 1 out of 16 consecutive channels can be used as digitizer at a time.

	Trigger Based	Vector Based
Max. Sample Rate	250ksps	40ksps
Sample Averaging	N/A (always 1)	available
Sample Memory (per waveform)	4K	1M
Sample Period Resolution	25ns	Vector period resolution
Trigger Delay	20.0 μ s $\pm$ 1 μ s	<1 μ s

Note

To be able to capture more than 4k samples, a result area must be defined for PPMU or BADDC. The result area is to be defined as shown in the following example:

```
RESULT_AREA_CONFIG_SET resultAreaConfigSet("voltageRamp");
resultAreaConfigSet.pin("pin_1").setDcResultAreaSize(8192);
```

The example above reserves the space for 8192 samples in the memory that is shared by the error map and the digitizer samples.